# Finite Element Modeling of Electrochemical Impedance Spectroscopy in All-Solid-State Batteries using FEniCSx

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All-solid-state thin-film batteries are key components for the development of autonomous integrated systems, where performance depends on complex charge transport and interface kinetics [1]. Electrochemical Impedance Spectroscopy (EIS) represents a non-destructive technique used to characterize these phenomena by decoupling the resistive and capacitive contributions of the solid electrolyte and its interfaces [2]. In this work, we implement a numerical framework within FEniCSx to simulate the EIS response of such systems based on the Poisson-Nernst-Planck (PNP) model. This model couples electrostatic potential and ionic concentrations, accounting for the formation of the electrical double layers at the electrode-electrolyte interfaces [3]. We utilize the Unified Form Language (UFL) to perform an automatic linearization of the non-linear coupled system around various electrode bias potentials. This approach allows for a direct resolution in the frequency domain. The central feature of the implementation is the post-processing of the frequency-domain results, using a residual-based current calculation to ensure strict electrical conservation. We demonstrate the framework's validity by comparing simulated Nyquist plots with experimental data obtained from thin -film batteries. This work provides a physically consistent tool for the study of electrochemical phenomena in solid-state energy storage.

## References

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